College Algebra Linear Inequalities in Two Variables

1) Graph the solution set of the linear inequality in two variables.

$y \geq 3x - 1$

a) Is (2, 3) a solution to the inequality $y \geq 3x - 1$?

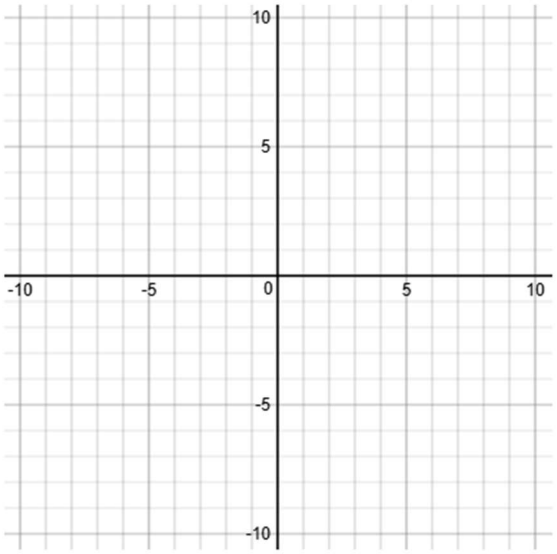
b) Is (1, 4) a solution to the inequality $y \geq 3x - 1$?

c) Graph the solution set of the inequality $y \geq 3x - 1$?

d) Is the boundary line solid or dashed?

e) What is the y-intercept of the boundary line?

f) Is the shading above or below the y-intercept?



2) Graph the solution set of the linear inequality in two variables.

$y \leq \frac{2}{3}x + 2$

a) Is (6, 3) a solution to the inequality $y \leq \frac{2}{3}x + 2$?

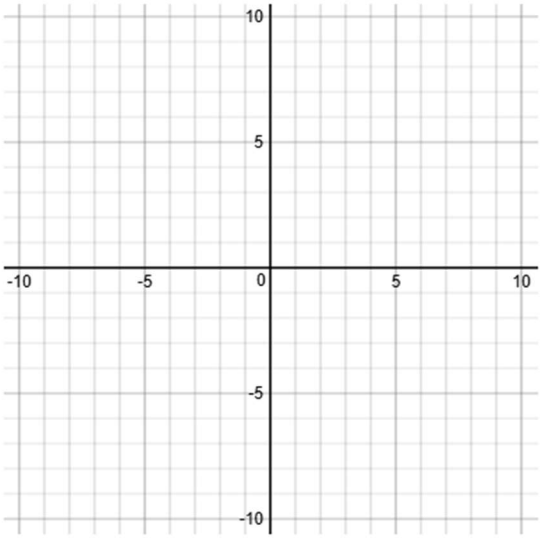
b) Is (1, 4) a solution to the inequality $y \leq \frac{2}{3}x + 2$?

c) Graph the solution set of the inequality $y \leq \frac{2}{3}x + 2$?

d) Is the boundary line solid or dashed?

e) What is the y-intercept of the boundary line?

f) Is the shading above or below the y-intercept?



3) Graph the solution set of the linear inequality in two variables.

$3x + y \geq 1$

a) Is (6, 3) a solution to the inequality $3x + y \geq 1$?

b) Is (1, -4) a solution to the inequality $3x + y \geq 1$?

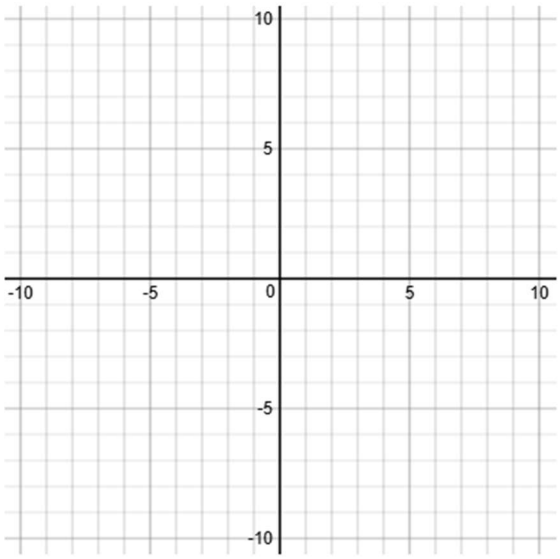
c) Solve the inequality $3x + y \geq 1$ for y. $y \geq$ _____?_____

d) Use the result from part (c) to graph the solution set of the inequality $3x + y \geq 1$.

e) Is the boundary line solid or dashed?

f) What is the y-intercept of the boundary line?

g) Is the shading above or below the y-intercept?



4) Graph the solution set of the linear inequality in two variables.

$-2x + 4y \geq -1$

a) Is (6, 3) a solution to the inequality $-2x + 4y \geq -1$?

b) Is (1, -4) a solution to the inequality $-2x + 4y \geq -1$?

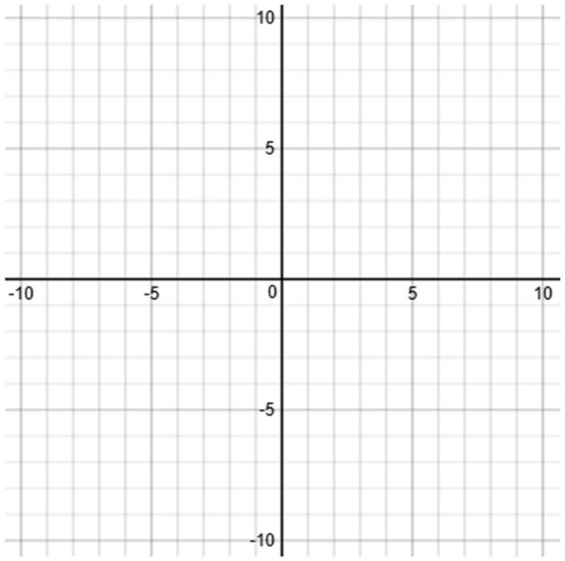
c) Solve the inequality $-2x + 4y \geq -1$ for y. $y \geq$ _____?_____

d) Use the result from part (c) to graph the solution set of the inequality $-2x + 4y \geq -1$.

e) Is the boundary line solid or dashed?

f) What is the y-intercept of the boundary line?

g) Is the shading above or below the y-intercept?



5) Graph the solution set of the linear inequality in two variables.

$3x - 4y \geq -12$

a) Is (6, 3) a solution to the inequality $3x - 4y \geq -12$?

b) Is (1, -4) a solution to the inequality $3x - 4y \geq -12$?

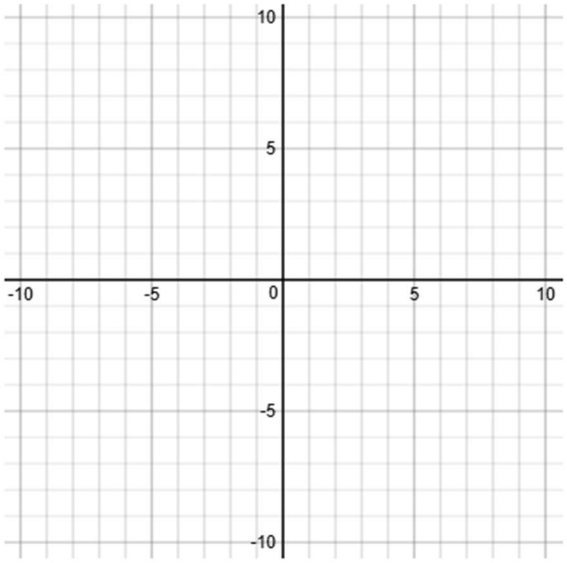
c) Solve the inequality $3x - 4y \geq -12$ for y. $y \leq$ _____?_____

d) Use the result from part (c) to graph the solution set of the inequality $3x - 4y \geq -12$.

e) Is the boundary line solid or dashed?

f) What is the y-intercept of the boundary line?

g) Is the shading above or below the y-intercept?



6) Graph the solution set of the linear inequality in two variables.

$2x - 5y < -10$

a) Is (6, 3) a solution to the inequality $2x - 5y < -10$?

b) Is (1, -4) a solution to the inequality $2x - 5y < -10$?

c) Solve the inequality $2x - 5y < -10$ for y. $y >$ _____?_____

d) Use the result from part (c) to graph the solution set of the inequality $2x - 5y < -10$.

e) Is the boundary line solid or dashed?

f) What is the y-intercept of the boundary line?

g) Is the shading above or below the y-intercept?

